

SECTION 7 – RF SHIELDED ROOM

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7-1 RF SHIELDED ROOM SPECIFICATION

The shielding effectiveness of the RF Shielded Room impacts the imaging ability of the system. Therefore, the RF Shielded Room needs to meet the attenuation specification before the upgrade. The RF Shielded Room RF attenuation specification with the GE system connected is 85 dB planewave (10MHz – 100MHz). If the RF Shield is not at this specification then RF vendor maintenance may be necessary.

If the magnet is upgraded to an LCC magnet with the Signa TwinSpeed system upgrade and the RF Shield is opened, modified, upgraded, etc. then the RF Shielded Room integrity **MUST** be revisited (tested) to meet or exceed the new system RF Shield Room specifications. Refer to ***Direction 2294003 Signa TwinSpeed Pre-Installation Section 7-1 RF SHIELDED ROOM SPECIFICATION.***

7-2 VENTS

If an upgrade to an LCC magnet is being performed with the Signa TwinSpeed system upgrade then refer to ***Direction 2294003 Signa TwinSpeed Pre-Installation Section 7-2 VENTS*** for cryogenic & HVAC vents requirements and recommendations.

7-3 PLUMBING

The Signa TwinSpeed system performance requires adherence to the specifications in ***Direction 2294003 Signa TwinSpeed Pre-Installation Section 7-3 PLUMBING*** for pipes, water, medical gasses, and sprinklers.

7-4 ELECTRICAL

The Signa TwinSpeed system performance requires adherence to the specifications in ***Direction 2294003 Signa TwinSpeed Pre-Installation Section 7-4 ELECTRICAL*** for filter attenuation, power line grounding, lighting, and common ground stud proximity.

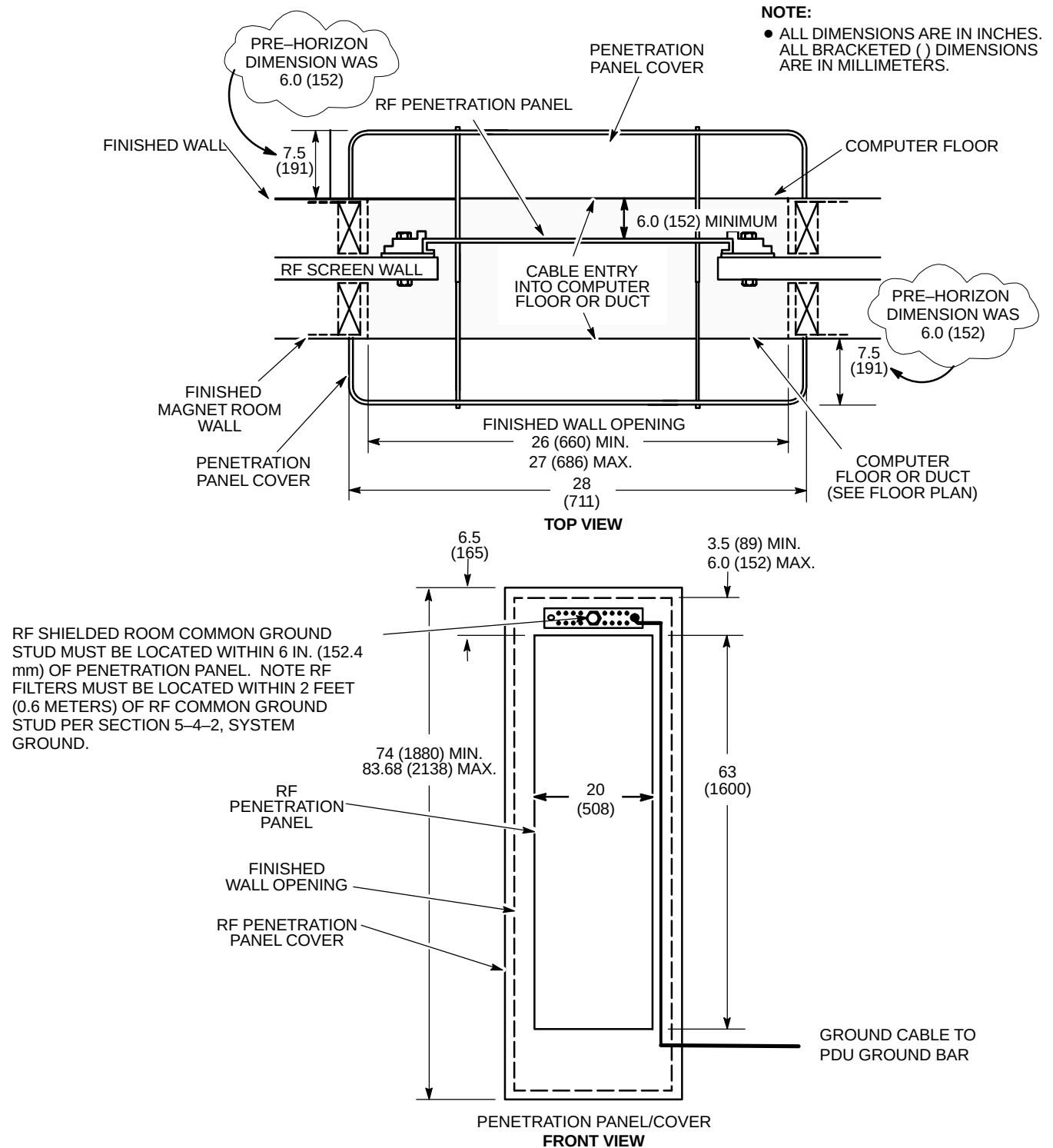
7-5 RF PENETRATION PANEL

The Penetration Panel is modified for Signa MR/i, CV/i, CV/i and LX upgrades. The Penetration Panel is replaced during Signa Horizon and prior system configurations upgrades as shown in Section 1 – SYSTEM CONFIGURATION. Current panel mounting hardware will be used to mount new Penetration Panel. The new Penetration Panel is deeper therefore new deeper Penetration Panel Covers are provided. The mounting and clearance dimensions for the Penetration Panel covering are shown in Illustration 7-1. Refer to Section 2-9, COMPONENT DIMENSIONS, for specific upgrade catalogs provide new Penetration Panel and Covers.

Note

The Penetration Panel must be covered on both sides for safety. If GE supplied adjustable covers are not used, customer furnished covers or enclosures with key or tool are required for opening to limit access to the panel.

7-5 RF PENETRATION PANEL



PENETRATION PANEL/COVERING MOUNTING REQUIREMENTS

ILLUSTRATION 7-1

7-6 PHYSICAL CONSIDERATIONS

If an upgrade to an LCC magnet is being performed with the Signa TwinSpeed system upgrade then refer to ***Direction 2294003 Signa TwinSpeed Pre-Installation*** following sections for **Section 7-6 PHYSICAL CONSIDERATIONS** for door and other openings, ceiling height, walls, and floors requirements and recommendations.

7-7 ANCHOR HARDWARE REQUIREMENTS FOR MR EQUIPMENT INSIDE RF SHIELDED ROOM

7-7-1 Magnet Mounting

If an upgrade to an LCC magnet is being performed with the Signa TwinSpeed system upgrade then refer to ***Direction 2294003 Signa TwinSpeed Pre-Installation*** following sections for **Section 7-7 ANCHOR HARDWARE REQUIREMENTS FOR MR EQUIPMENT INSIDE RF SHIELDED ROOM** for Magnet and Dock Assembly requirements.

7-7-2 Blower Box

The Blower Box (MG6) provides cooling air for the Patient Comfort Module in the Magnet Enclosure for Signa TwinSpeed system.

The existing system Blower Box Panel is modified (cover plate installed) for upgrades to Signa MR/i, CV/i, CV/i, LX and CVMR with LCC Magnet installed so the existing box mounting will be used.

A new Blower Box is installed by Signa LX and CVMR without LCC Magnet installed, Signa Horizon and prior system configurations upgrades. The Blower Box will be mounted within the RF Shielded Room and connect to the Patient Comfort Module by 4.5 inch (114.3 mm) OD flexible vinyl air ducting. **The Blower Box contains magnetic material and must be securely mounted on the floor of the Magnet Room, a raised support shelf, or the Magnet Room wall or ceiling with support provided under box.** RF Shield integrity must be maintained for mounting the Blower Box within the Magnet Room. The flexible vinyl air duct route is routed from the Blower Box through the Magnet Enclosure Rear Pedestal cable access and connect to the Gradient Coil. Refer to Illustrations 2-33 for Blower Box and mounting dimensions. Refer to ***Direction 2294003 Signa TwinSpeed Pre-Installation*** following sections for **Section 7-7 ANCHOR HARDWARE REQUIREMENTS FOR MR EQUIPMENT INSIDE RF SHIELDED ROOM** for Blower Box mounting RF Shield Integrity and electrical isolation requirements.

7-8 MAGNET ROOM EQUIPMENT MOUNTING**7-8-1 Magnet Rundown Unit (MS4)**

If an upgrade to an LCC magnet is being performed with the Signa TwinSpeed system upgrade then the magnet rundown unit should be mounted 60 in. (1524 mm) above the Magnet Room floor near the front of the Magnet Room but outside the 200 gauss zone.

7-8-2 Emergency Off Buttons

Emergency off buttons to be located near the exit in the magnet and equipment rooms. These buttons must be clearly labeled, "Emergency Off". Refer to Section 5, POWER REQUIREMENTS, for emergency off power requirements.

7-9 ROOM VENTILATION SWITCH

Placement of the room ventilation switch should be near the Magnet Room door and is the responsibility of the architect and mechanical contractor.